



Valentinus Mahendra Aaron Quendangen

Jakarta, Indonesia

aaronvalentinus@gmail.com • +62 831-1460-2668

<https://ronaaron61.github.io/#/about>

Motivated to work hard in carrying out work activities to achieve success and gain experience, such as learning new things to gain more experience, especially by doing or helping with a project or research.

Education

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| • SMAK Santo Yoseph Denpasar
Science Major | 2016 - 2019 |
| • Universitas Airlangga
Biomedical Engineering – Medical Instrumentation Major
GPA/IPK: 3.68 | 2019 - 2023 |

Experiences

Research Assistant Project

Jakarta

Indonesian Medical Education and Research Institute (IMERI) FKUI

May 2026 - Present

- Developed and submitted research proposals, including budgets, methodologies, and timelines
- Acted as a liaison between doctors, specialists, and external partners.
- Supervised interns, led site visits, and served as a speaker in academic sessions.
- Assisted in research data collection and verification.

IPSRs – Medical Equipment Maintenance

Pontianak

Rumah Sakit Umum Medika Djaya

November 2025 – May 2026

- Weekly and monthly checks on medical equipment in the ICU, ward, physiotherapy, emergency department, and the medical gas area.
- Conduct troubleshooting, calibration, and repair of faulty medical equipment.
- Support the installation of new equipment and conduct accompanying user training.
 - Maintenance, Problem Solving, Communication, Electricity, Soldering, Electromedic

ENT Product Specialist

Surabaya

PT Edison Duta Sarana

July 2024 – October 2024

- Manage the supply and stock of ENT medical devices and instruments.
- Communicating between principals, purchasing, marketing, and doctors about the products needed, including instruments, ENT Unit, and endoscopy camera.
- Training new users on using the product, assisting in the use of the medical equipment during surgery, and troubleshooting product problems.
- Planning and creating product lists, specifications, price quotes, and presentations
 - Speaking (Indonesian and English), Communication, ENT basic, Management, Problem Solving

Research Engineer

Surabaya

PT Medika Karya Airlangga

December 2023 – November 2025 (*Full Time*) – Present (*Part Time*)

- Research and development of a smart phototherapy system for jaundiced babies - Smart Phototherapy System Airlangga Bilirubin Nesting (AirBiliNest) Model 2 and 3. Including prototyping the electrical components, design, programming, application, and data gathering.
- Charged with managing technical issues, including prototyping, PCB, wiring design, and assembly of the controller, lamp design, sensor development, microcontroller programming, technical drawings, Android application development, and web development.

- Gather safety and efficacy data in Airlangga University Hospital, and process the data for the paper and future development.
 - Troubleshooting, basic electronics, soldering, Microcontroller, PCB design, Graphic Design, Laser Cutting, Soldering, Communication, Testing, Demonstrating, RnD, Android Studio, 3D Printing Design, front-end Web Design, back-end web design, Autodesk Inventor

Research Engineer Assistant

Surabaya

Neonatal Research Group FK Universitas Airlangga

August 2023 - Present

- Develop the research of Smart Phototherapy System Airlangga Bilirubin Nesting (AirBiliNest), which is a smart phototherapy device for jaundiced babies that can be monitored and controlled using an Android application. Assembling, testing, and multiplying the prototype required.
- Gather safety and efficacy data in Airlangga University Hospital, and process the data for the paper and future development.
- Implementing the MBKM Matching Fund events related to this device with the students, faculty, and industry partners in West Java.
 - Troubleshooting, Microcontroller, Laser Cutting, Soldering, Communication, Testing, Demonstrating

Merdeka Belajar Kampus Merdeka (MBKM) Matching Fund

Surabaya

PT Integrasi Bisnis Eksekutif (IBE Reality)

September 2022 - January 2023

- Assisted with the manufacturing of X-Mano, the Wearable Exoskeleton Hand Robot for Movement Therapy of Post-Stroke Patients, through the 4-month internship. Including the 3D design for the case, microcontroller, assembly, and Android application.
- Assisted the clinical trial to gain feedback from 20+ post-stroke patients in Universitas Airlangga Hospital.
 - 3D Printing, Autodesk Inventor, Microcontroller, Clinical trial, Entrepreneurship

Mentor on Coding Workshop for Biomedical Engineering

Surabaya

Himpunan Mahasiswa Teknik Biomedis (HMTB) UNAIR

March 2021 - June 2021

- Teach the basic use of syntax, functions, and how to make simple programs using C++ and Python for the Biomedical Engineering students class of 2021.
- Check and evaluate the results of the program creation task.
 - C++, Python, Teaching, Speaking

Technical Expertise

Programming (Python/MicroPython, C++/Arduino, Matlab, HTML) | Arduino, ESP, Analog Discovery | Basic electronics, Electrical circuits, analog and digital electronics, Soldering, PCB design, 3D Printing, Laser cutting, Graphic design, Autodesk Inventor/solidworks.

Achievements

- Best teamwork at Sebelas Maret International IoT Challenge 2021 – Smart Trash Bin to prevent COVID-19
- Silver award for iMIT SIC 2021 (4th International Malaysia-Indonesia-Thailand Symposium on Innovation and Creativity 2021) held by UiTM Cawangan Malaysia, “3d (Three-Dimensional) Lung Cancer Detection and Identification Based On Mixed Reality”

Research and Publications

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| • Edukasi Siswa Sma Negeri 1 Kampak Kabupaten Trenggalek Provinsi Jawa Timur Melalui Pelatihan Rancang Bangun Alat Deteksi Kelelahan Berbasis Audiovisual Dalam Peningkatan Mutu Kerja Dan Kesehatan. (Vol. 4 No. 2: Jurnal Pepadu, DOI:10.29303/pepadu.v4i2.2370) | 2023 |
| • Design Of Multifrequency Electrical Impedance Tomography (Mfeit) Based On Analog Discovery To Detect Breast Cancer. (Vol. 87 No. 1: Jurnal Teknologi, DOI:10.11113/jurnalteknologi.v87.21921) | 2024 |
| • Evaluation of New Portable Phototherapy Device for Neonatal Hyperbilirubinemia (International Research Journal of Multidisciplinary Scope (IRJMS), DOI:10.47857/irjms.2025.v06i02.03766) | 2025 |

Mini Projects

Projects: <https://ronaaron61.github.io/#/projects>

- Electrical Impedance Tomography using ESP Microcontroller to detect anomalies in a phantom that looks like a human body
- Bioimpedance measure using a Microcontroller, and can be used to detect hand angle estimation to control the mechanical arm
- Simple EMG and ECG module for a microcontroller using AD620
- Virtual mouse and keyboard that can be controlled by finger position in the webcam using Python and the mediapipe library
- A smart trash bin that can be opened by gently kicking the bin or placing a hand near the sensor. Can be connected to the Blynk app to monitor the trash volume